

PROFESSIONAL SUMMARY

Software Engineer with 6+ years of experience specializing in **C++, Qt/QML, V8 JavaScript Engine and cross-platform development**. Proven expertise in building **secure SDKs, real-time systems,, and scalable backend architectures**. Strong background in **game development, software engineering, and distributed systems**. Experienced in delivering high-performance applications across Windows, Linux, macOS, Android, and iOS.

TECHNICAL SKILLS

- **Languages:** C/C++ (Primary), Java, JavaScript (ES6), C#, SQL
 - **Frameworks/Tools:** Qt/QML, Node.js, Chrome V8 Engine, SWIG, Boost
 - **Backend:** REST APIs, gRPC, SignalR, WebSockets, Microservices, MVC
 - **Databases:** MySQL, PostgreSQL, Redis
 - **Systems:** Cross-platform development, Distributed systems, Clustering
 - **Protocols:** HTTP/HTTPS, WebSocket, gRPC, FTP
 - **Embedded:** UART, SPI, I2C, Microcontrollers (AVR)
 - **Version Control:** Git, SVN
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PROFESSIONAL EXPERIENCE

C++ Software Engineering (Qt/QML) /(Java/C#) — Softlock

Jan 2022 – Present

- Developed and maintained in a collaboration with AET Europe a language-agnostic, cross-platform SDK for authentication and digital signing (eIDAS compliant).
- Developed a high-performance C++/Qt core for the former SDK, resolving intricate design challenges, multithreading issues, and memory leaks across Windows, Linux, and macOS. Additionally, created wrappers using JNI and SWIG for platform integration.
- Developed a desktop admin tool in C++/Qt/QML for security tokens, facilitating the generation of public and private key pairs and managing encryption processes.
- Played a pivotal role in the development of Windows drivers for encryption and decryption, contributing significantly to the security and functionality of projects.
- Spearheaded the development of a successful cross-platform mobile application supporting FIDO2 authentication, integrating NFC hardware authentication. Also, wrote native code in Java and Objective-C for seamless library interfacing.
- Engaged in the complete life cycle of an ASP.NET web project, from design to implementation and deployment in a Kubernetes cluster.
- Build Credential Provider for windows login System to support Fido2 Devices and OTP offline login
- Build PAM tool for Linux to enable authentication using FIDO2 Smartcard

C++ Software Engineering – ViaStore (Part Time)

October 2023 – Jun 2024

- Developed and maintained C++14 code for a comprehensive Warehouse Management System (WMS), focusing on enhancing automation and efficiency.
- Implemented new features in C++, analyzed a lot of legacy code, and optimized system performance through improved coding
- Practice and algorithm optimization.
- Utilized Qt for developing user interfaces and SQL databases for managing data, ensuring robust and scalable solutions.
- Analyzed database structure and created effective SQL requests to streamline data operations.
- Worked with SVN for version control and successfully converted FTP to FTPS to enhance data security

Game Developer (C++ / Unreal Engine)

2018 – 2022

- Developed **real-time multiplayer and RTS games** using Unreal Engine (C++).
 - Implemented networking systems, gameplay mechanics, and performance optimizations.
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EDUCATION

Bachelor of Computer Engineering — Benha University (2015 – 2020)

Grade: Very Good | Graduation Project: Excellent

IELTS: B2

PROJECTS

1. **Vehicle Collision Detection System:** Real-time vehicle communication and collision prediction system (Graduation Project – Excellent).
2. **C++/Java/C# SDK** for remote signing and smartcard signing (Private bitbucket)
3. **Developed and maintained** C++14 code for a comprehensive Warehouse Management System (WMS), focusing on enhancing (Private SVN)
4. **Port cocos2d-js** from Javascript to Typescript and convert renderer from canvas to WebGL2
5. **Port cocos2d-x** from Spidermonkey JS Scripting Engine to V8 Engine
6. **Credential provider** for windows login Using Fido2 Smartcard using libfido2
7. **Credential provider** for windows to enable login using offline OTP using Smartcard and TPM
8. **Qt Server** using QtHttp/QtWebsocket and Database driver to build backend for online game
9. **Linux PAM** using Fido2 Device implemented in libfido2
10. **Massive Online Game (C++ / Cocos2d):** Scalable multiplayer game engine.
11. **Cross-Platform Game (JS + Node.js):** Multi-platform architecture with real-time communication.
12. **Online RTS Game (Elkaiser):** Built with native web stack and Unreal Engine.
13. **Q&A Platform:** Full-stack web application.
14. **APK Store Platform:** Managed multi-language platform with 6M+ apps.